

DIYA KINI

Austin, TX | (512) 944-3923 | [Email](#) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

University of Michigan

Ann Arbor, MI

B.S.E in Computer Science, Business Minor

2023 - 2026

- Academic Honors: Michigander Scholar, Dean's Honor List, Member of Amazon's Women of the World
- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Discrete Mathematics, Computer Vision

EXPERIENCE

Lockheed Martin (LAIC)

Remote

Incoming AI Intern

May 2025 - July 2025

University of Michigan Robotics – The MBot Ecosystem

Ann Arbor, MI

Research Assistant

Jan 2024 - May 2024

- Led autonomous robotic system development and improvement for 100+ vision-controlled robots
- Applied mathematical operations to Python, OpenCV, and computer vision algorithms to facilitate distance learning
- Enhanced the Jetson Nano's stereo vision implementation within the MBot ecosystem through ORB SLAM
- Integrated the updated vision systems in 2 University of Michigan's robotics courses, impacting 40 graduate students

University of Michigan College of Engineering

Ann Arbor, MI

Computational Linear Algebra Instructional Aide

Jan 2024 - Present

- Executed matrix operations and linear transformations to optimize LiDAR data interpretations in robotic navigation
- Taught 200+ students the application of ML concepts, including linear regression, to develop predictive models for robotic systems, enhancing pattern recognition and decision-making processes
- Explained feedback control algorithms for dynamical systems, such as balancing a Segway robot, leveraging linear algebra and Julia programming for improved system stability and performance

Berry Consultants, LLC

Austin, TX

Data Science Intern

June 2022 - Aug 2022

- Analyzed the optimal length of time a novel gene therapy would keep hemophilia patients at a safe level of clotting
- Worked with a team of 3 data scientists to investigate the relationship between biomarkers of the gut biome and clinical outcomes for an innovative treatment for C-difficile infections through the R programming language
- Assisted in the FDA approval of 2 treatments by developing plots of drug efficiency and safety

PROJECTS

CV Fluid Dispensers (CV & Robotics)

Feb 2025

- Collaborated within an agile environment to create a multi-dispensable product for use in minimal contact spaces
- Led computer vision aspects for motion and hand signal detection through webcam and raspberry pi implementation
- Designed and manufactured components using 3D printing and CNC, optimizing materials for compartment pressure

Predicting FX Rates with Sentiment Analysis (TensorFlow)

July 2024

- Used time-series analysis and LSTMs to predict EUR/USD exchange rate from existing data in Yahoo Finance
- Incorporated news sentiments from past news articles to improve forecast accuracy through NLP techniques and Tensorflow by analyzing the emotional tone of words in the text discussing historical conflicts between nations

ML Robot Tour (Python)

Dec 2023

- Integrated vision-based navigation system, enabling adaptive robot pathfinding upon numerical cue detection
- Optimized route selection of MBot Omni's obstacle avoidance by an increase of 22% through SLAM and BFS
- Trained 3 ML numerical recognition models (KNN, Linear Classifier, NN) using SciKit Learn and MNIST database

SKILLS

Licenses & Certifications:

TensorFlow Professional Developer Certificate (DeepLearning.AI),
Quantitative Modeling for Analysts Specialization (Wharton Online)

Programming Languages & Computer Skills:

Python, C++, R, Java, Solidworks, Julia, OpenCV, ROS, MATLAB, Verilog,
Git Version Control

Tools & Machinery:

CNC, Mill, Lathe

Extra skills:

Spanish, Leadership, Public Speaking